

The CFAC Theorem:

Doi–Peliti Generating Functions Are Lagrange Equations

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Abstract

A chemical reaction network (CRN) of physical interest typically lives in a continuous fluctuating medium: bacteria in a chemoattractant, protein monomers in a cytoplasmic density field, ice nucleation sites in supersaturated vapour. Treating the discrete reacting species with Doi–Peliti (DP) and the continuous medium with Martin–Siggia–Rose (MSR) produces a coupled DP–MSR action whose vertex content is polynomial. We show that for any such coupled theory the dressed-propagator generating function satisfies a Lagrange equation in the sense of Flajolet–Sedgewick analytic combinatorics. The recognition is an elementary rewriting; the consequences are the transfer theorem for all-orders coefficient asymptotics, classification of mean-field universality by Puiseux exponent, a branch-gap contour integral for non-perturbative nucleation rates, and closed algebraic form at one loop via tropical geometry. We call the resulting framework Coupled Field Analytic Combinatorics (CFAC). This note states the theorem, gives the proof, and lists what can be read off from any CRN-in-medium once written in DP–MSR form.

1 Definitions

CRN in a continuous medium. A reaction network of physical interest rarely sits in isolation; it is embedded in a continuous medium whose fluctuations couple back to the reaction kinetics. Bacteria secrete and consume a chemical field, amyloid monomers diffuse in a thermally fluctuating cytoplasm, nucleation sites see a supersaturated carrier. Pruessner and Garcia-Millan ([Pruessner and Garcia-Millan, 2025](#)) and Bothe *et al.* ([Bothe et al., 2023](#)) have recently emphasised that coarse-graining the discrete reacting species into a pure continuous-field description produces *spurious* entropy production: the particle sector must be retained alongside the field sector. CFAC takes this as given and treats both sectors on equal footing: discrete reacting species via Doi–Peliti ([Doi, 1976](#); [Peliti, 1985](#); [Täuber, 2014](#)), the continuous medium via Martin–Siggia–Rose ([Martin et al., 1973](#); [Janssen, 1976](#); [De Dominicis, 1976](#)), with interactions between them as coupling vertices. The rest of this section sets up the object that results: a Liouvillian with a polynomial vertex dictionary, equally at home on the particle legs and the field legs.

Definition 1 (Chemical reaction network in a continuous medium). A CRN-in-medium is a finite collection of discrete reacting species $\{A_1, \dots, A_s\}$ together with reactions $\{\sum_i k_{ri} A_i \rightarrow \sum_i l_{ri} A_i\}_r$ with nonnegative integer stoichiometry and rates $k_r > 0$, and optionally a finite collection of continuous medium fields $\{\phi_1, \dots, \phi_m\}$ with stochastic dynamics $\partial_t \phi_j = \mathcal{F}_j(\{\phi\}, \{A_i\}) + \eta_j$, η_j Gaussian noise. Coupling vertices $A_i \longleftrightarrow \phi_j$ may appear in both the reaction rates and $\mathcal{F}_j$. When no medium is present ($m = 0$) this reduces to a standard CRN.

Definition 2 (Liouvillian of a CRN-in-medium). For each discrete species A_i introduce Heisenberg–Weyl creation and annihilation variables $\tilde{\psi}_i, \psi_i$; for each continuous medium field ϕ_j introduce the MSR response/field pair $\tilde{\phi}_j, \phi_j$. To each reaction r associate the operator

$$\hat{Q}_r = k_r \left[\prod_i \tilde{\psi}_i^{l_{ri}} - \prod_i \tilde{\psi}_i^{k_{ri}} \right] \prod_i \psi_i^{k_{ri}} \quad (1)$$

in the “ordinary” (non-Doi-shifted) form, or equivalently

$$Q_r = k_r \left[\prod_i (1 + \tilde{\psi}_i)^{l_{ri}} - \prod_i (1 + \tilde{\psi}_i)^{k_{ri}} \right] \prod_i \psi_i^{k_{ri}} \quad (2)$$

in the Doi-shifted form. The MSR action for each medium field is $S_j^{\text{MSR}} = \int dt d\mathbf{x} [\tilde{\phi}_j(\partial_t \phi_j - \mathcal{F}_j) - \frac{1}{2} D_j \tilde{\phi}_j^2]$. The total Liouvillian is $Q = \sum_r Q_r + \sum_j \mathcal{L}_j^{\text{MSR}}$ together with coupling vertices $A_i \leftrightarrow \phi_j$ read off from the reaction rates that depend on ϕ and from the ϕ_j -equation terms that depend on $\{A_i\}$. For $m = 0$ (no medium fields) this reduces to the pure Doi–Peliti Liouvillian.

The two forms (1)–(2) are related by the Doi shift $\tilde{\psi}_i \rightarrow 1 + \tilde{\psi}_i$ (Doi, 1976; Peliti, 1985; Täuber, 2014). Expanding any product $\prod_i (1 + \tilde{\psi}_i)^{l_{ri}}$ or $\prod_i \tilde{\psi}_i^{l_{ri}}$ and collecting like terms gives a *vertex dictionary*:

$$Q = \sum_{(m_1+n_1 \geq 1, \dots)} g_{\{m\}, \{n\}} \prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}, \quad (3)$$

with coupling coefficients $g_{\{m\}, \{n\}}$ determined algorithmically by the stoichiometry and rates (Amarteifio, 2019). Each term $\prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ is an interaction vertex; its n_∂ value is zero unless spatial gradients are added by hand (e.g. to model drift or chemotaxis).

Definition 3 (Polynomial vertex structure). A CRN has *polynomial vertex structure* if the Liouvillian, after the Doi shift (2) and possibly the projection to zero spatial dimensions (replacing $\psi_i(\mathbf{x}, t) \rightarrow \psi_i(t)$), expands into a finite sum of monomials $g_{\{m\}, \{n\}} \prod_i \tilde{\psi}_i^{m_i} \psi_i^{n_i}$ with no momentum-dependent couplings.

Every CRN with finite stoichiometry has polynomial vertex structure in this sense. Continuous gradient vertices (e.g. the chemotaxis term $\chi \tilde{\psi}(\nabla \phi)(\nabla \psi)$) are *not* of this form at nonzero momentum; their zero-momentum / well-mixed projections are.

Definition 4 (Dressed-propagator generating function). Fix a species A with self-coupling vertices only (multi-species cases are handled by resultants; see §7). The dressed propagator is $G(z) = \sum_{n \geq 0} G_n z^n$, where G_n is the sum over connected 1PI diagrams contributing to the two-point function at order n in the bare propagator z . The *vertex function* is

$$\phi(G) = 1 + \sum_{m+n \geq 3} g_{mn} G^{m+n-2}, \quad (4)$$

summing over all vertices $g_{mn} \tilde{\psi}^m \psi^n$ of the Liouvillian with $m+n \geq 3$. (Two-leg vertices $m+n = 2$ are mass terms absorbed into z .)

2 The CFAC theorem

Theorem 1 (CFAC recognition). *For any CRN with polynomial vertex structure, the dressed-propagator generating function $G(z)$ and the vertex function $\phi(G)$ of Eq. (4) satisfy*

$$\boxed{G(z) = z \cdot \phi(G(z))} \quad (5)$$

as a closed functional equation over formal power series in z . Equation (5) is a Lagrange equation in the sense of analytic combinatorics (Flajolet and Sedgewick, 2009).

Consequently, for any such CRN the following quantities are determined directly by the polynomial ϕ :

(C1) All-orders coefficient asymptotics. If z_* is the radius of convergence of $G(z)$, with Puiseux exponent p/q at z_* , then by the transfer theorem

$$[z^n] G(z) \sim \frac{C}{\Gamma(-p/q)} n^{-p/q-1} z_*^{-n}, \quad n \rightarrow \infty, \quad (6)$$

with C set by the local behaviour of ϕ near (G_*, z_*) .

(C2) Mean-field universality class. *The rational Puiseux exponent p/q at the dominant branch point of (5) classifies the mean-field scaling: $1/2$ is the directed-percolation / pair-annihilation class, $1/3$ is the compact-DP / triple-root class, higher-order mergers give further multi-critical classes.*

(C3) Non-perturbative nucleation barrier. *When ϕ produces a bistable effective dynamics, the nucleation barrier is the contour integral*

$$S = \int_1^{z^*} \frac{G_{\text{unst}}(z) - G_{\text{stab}}(z)}{z} dz, \quad (7)$$

equal to the Freidlin–Wentzell / WKB quasi-potential $\int_{x_1}^{x_2} \ln[w^-/w^+] dy$ by change of variables.

(C4) UV divergence structure. *At spatial dimension d , the superficial degree of divergence of a 1PI graph with L loops, I internal lines, and n_g^{tot} gradient insertions is $\omega = dL - 2I + n_g^{\text{tot}}$. Power counting on the finite enumeration of 1PI graphs consistent with the vertex dictionary is then purely algebraic.*

Scope. The theorem as stated holds when ϕ is a polynomial in G alone. This covers every CRN with finite stoichiometry at zero spatial dimensions or after integrating out additional field variables at steady state. When gradient vertices carry momentum-dependent couplings, ϕ becomes a functional of momentum-dependent propagators; the recognition extends, but the transfer-theorem machinery becomes functional-analytic. All applications in this note stay within the polynomial scope.

Corollary 2 (CFAC pipeline). *For any CRN with polynomial vertex structure, the following chain is purely algebraic and mechanical:*

stoichiometry $\rightarrow$ Liouvillian $Q \rightarrow$ vertex dictionary $\{g_{mn}\} \rightarrow$ vertex function $\phi(G) \rightarrow$ Lagrange equation $G = z\phi(G) \rightarrow$ (C1) asymptotics, (C2) universality, (C3) nucleation, (C4) divergences.

No loop integral, no saddle-point of an action, no WKB trajectory is required to reach any of (C1)–(C4).

Corollary 3 (Rare-event scaling). *For a bistable CRN with polynomial vertex structure and effective system size V , the mean first-passage time from the metastable state to the unstable barrier is*

$$\tau(V) = C e^{VS} [1 + O(1/V)], \quad (8)$$

with barrier S given by the branch-gap integral (7) of $\phi(G)$ and prefactor C determined by the local Puiseux data at the branch point via the transfer theorem (6). Neither direct stochastic simulation nor action saddle-pointing is required to extract S .

Proof. Equation (8) is the large-deviation form of the Freidlin–Wentzell escape rate (Freidlin and Wentzell, 2012). By (C3), S equals the branch-gap integral of the Lagrange equation (5). The prefactor C comes from the Gaussian fluctuations around the instanton trajectory; in the Lagrange-equation language these fluctuations are controlled by the local coefficients of the Puiseux expansion at (G_*, z_*) , which feed into (6). The two pieces of data (S, C) are thus both read off from the polynomial ϕ without simulation or WKB. The companion paper (Amarteifio, 2026b) carries this out explicitly for the Schlögl-I model, giving S and C_∞ as closed-form algebraic functions of the rate constants (via the branch-point biquadratic and a Kramers-type formula built on the same polynomial data), verified against Gillespie simulation. $\square$

Corollary 4 (Coupled field extension via resultant). *For a CRN on s species with polynomial vertex structure across all species, the system of Dyson–Schwinger equations $G_i = z_i \phi_i(\{G_j\})$ for $i = 1, \dots, s$ admits a univariate reduction: the resultant $\mathcal{R}(G_1, z_1; \{z_j\}_{j \neq 1})$ obtained by eliminating $G_2, \dots, G_s$ from the system is a polynomial in G_1 whose roots $G_1(z_1)$ inherit a Lagrange-equation structure. Theorem 1 applies to the resultant with the following scope:*

- *Consequences (C1), (C2), (C4)—coefficient asymptotics, Puiseux classification of mean-field universality, and UV divergence structure—hold with ϕ replaced by $\mathcal{R}$, directly and unconditionally.*
- *Consequence (C3)—the branch-gap instanton integral—holds when the physical nucleation saddle is a real branch point of $\mathcal{R}$. It can fail when the saddle migrates into the complex analytic continuation of the resultant.*

Proof sketch and scope. The resultant of a polynomial system is itself polynomial in the retained variable (Cox et al., 2007); (C1), (C2), (C4) apply mechanically. For (C3), a numerical test on a 3-site coupled ring reported in (Amarteifio, 2026b) shows that the branch-gap formula on the resultant agrees with direct simulation in the decoupled limit (2.6%), under-predicts the barrier by $\sim 20\%$ in a weakly coupled regime, and has no real branch point at all in a moderately coupled regime, even though the simulated nucleation rate remains exponential. The nucleation saddle has left the real algebraic variety. Establishing (C3) in that regime requires complex-analytic continuation of the resultant, outside the polynomial-vertex scope of Theorem 1. $\square$

Corollary 5 (One-loop algebraic bridge via tropical geometry). *For any one-loop two-propagator integral arising in a CRN with polynomial vertex structure, the bridge integral (C4) admits a closed algebraic form: the Arkani-Hamed–Hillman–Salvatori leading-facet-plus-Hessian evaluation of the Schwinger representation is exact, not asymptotic. Explicitly, with first Symanzik polynomial $F = m_1^2 u_1^2 + (p^2 + m_1^2 + m_2^2) u_1 u_2 + m_2^2 u_2^2$,*

$$I_{\text{DP}}(p; m_1, m_2) = \frac{1}{\sqrt{\lambda_E}} \ln \left[\frac{(\sqrt{\lambda_E} + p^2 + m_1^2 + m_2^2)^2}{4 m_1^2 m_2^2} \right], \quad (9)$$

with $\lambda_E = (p^2 + m_1^2 - m_2^2)^2 + 4p^2 m_2^2$ the Euclidean Källén discriminant (equivalently, the determinant of the tropical Hessian at the interior saddle). The branch cut $\lambda_E = 0$ locates the Minkowski threshold $s = (m_1 + m_2)^2$ directly on the Newton polytope.

Proof sketch. F is homogeneous of degree $L+1 = 2$ in the Schwinger parameters; on the projective simplex it is a non-degenerate quadratic form with constant Hessian. The Schwinger integral of $F^{-d/2}$ is Gaussian against the leading-facet saddle, evaluated exactly by the saddle-plus-Hessian prescription (Arkani-Hamed et al., 2022; Borinsky, 2020): no higher Taylor corrections arise because $\ln F$ terminates at second order around the saddle up to a total derivative. Details, numerical verification across 14 orders of magnitude in p^2/m^2 and m_1^2/m_2^2 , and the extension to the coupled DP–MSR bridge appear in (Amarteifio, 2026b). $\square$ $\square$

Corollary 5 sharpens (C4): at one loop, the bridge step of the CFAC pipeline (Corollary 2) is not a numerical dictionary but a closed algebraic expression. Combined with the recognition theorem, the consequence is that for any CRN with polynomial vertex structure, the entire one-loop beta function is determined by symbolic manipulation of two polynomials: $\phi(G)$ (from the vertex dictionary) and F (the first Symanzik of the relevant topology). Higher loops require genuine multi-facet sums over Newton polytopes, for which Borinsky’s tropical Monte Carlo (Borinsky, 2020; Borinsky et al., 2025) gives an exact integration procedure.

Why one loop is often enough. The restriction of Corollary 5 to one loop is less severe than it appears, because (C4) itself is typically *one-loop-exact* for CRNs of interest. A CRN is one-loop exact when the ω -enumeration gives $\omega(L) < 0$ for every topology with $L \geq 2$ —meaning no higher-loop UV divergences arise. This is the situation for pair annihilation, pure DP, BARW, chemotaxis (Gelinson and Golestanian, 2015), the KS Doi–Peliti/MSR coupling sector (Pruessner and Garcia-Millan, 2025; Amarteifio, 2026b), and more generally for any CRN whose vertex content is sufficiently constrained by finite stoichiometry that the enumeration collapses. In those cases Corollary 5 and (C4) combine to close the *entire* perturbative pipeline in symbolic form: reaction

network $\rightarrow$ vertex dictionary $\rightarrow$ Lagrange equation $\rightarrow$ Puiseux exponent + branch-gap + closed-form bridge integral, with no numerical quadrature anywhere. The restricted class where this fails (e.g. active-birth ψ^4 sectors with persistent two-loop divergences) is exactly the class where ε -expansion resummation is also required in the standard direct-RG route; CFAC neither helps nor hurts there, and Borinsky’s algorithm remains the available exact evaluator.

What these corollaries are (and are not). Corollary 3 is a statement about the large- V form of the MFPT, using the standard large-deviation structure (Freidlin and Wentzell, 2012). The novelty is that *both* the exponent S and the prefactor C are read off from ϕ ; the present paper gives S rigorously and notes that C is in-principle determined by the transfer theorem. Corollary 4 is the technical observation that polynomial vertex structure across multiple species admits a univariate reduction by resultant, to which Theorem 1 applies unconditionally for (C1), (C2), (C4). For the nucleation consequence (C3), the reduction holds only when the physical saddle is a real branch point of the resultant; a 3-site lattice test in (Amarteifio, 2026b) finds this fails at moderate coupling, where the saddle is complex. The companion paper extends (C3) to that regime concretely by complex analytic continuation of the resultant with a Coleman–Callan sum over complex-conjugate saddle pairs, verified against Gillespie simulation to $\sim 3\%$ at $\alpha = 0.3$; the residual open problem is the general enumerability of reachable conjugate pairs from the real axis (a Galois/monodromy question on the resultant’s Riemann surface).

3 Proof

Proof of Theorem 1. The Dyson–Schwinger equation for the dressed propagator of a field theory with vertex content $\{g_{mn}\tilde{\psi}^m\psi^n\}$ is obtained by summing 1PI insertions into the bare propagator: $G = z + z \Sigma(G) G$, where the self-energy Σ is a sum over 1PI vertex insertions. A 1PI insertion of the vertex $g_{mn}\tilde{\psi}^m\psi^n$ contracts $(m-1)$ of its $\tilde{\psi}$ -legs and n of its ψ -legs to internal dressed propagators, contributing $g_{mn} G^{m+n-2}$. Summing these plus the identity gives (4). Substituting into the DSE:

$$G = z + z \Sigma(G) G = z[1 + \Sigma(G) G] = z \phi(G),$$

which is (5). The identification of this as a Lagrange equation over formal power series is definitional (Flajolet and Sedgewick, 2009, Ch. VII). Consequences (C1)–(C4) then follow directly: (C1) is the transfer theorem; (C2) is the Newton-polygon analysis of (5) at the discriminant zero; (C3) is proved by change of variables $y \rightarrow z = w^-(y)/w^+(y)$ in the quasi-potential integral; (C4) is Weinberg’s convergence theorem (Weinberg, 1960; Zimmermann, 1969) applied to the 1PI graph list produced by the algebraic formalism (Amarteifio, 2019), with the subdiagram condition automatic when the vertex dictionary is closed under 1PI restriction. $\square$

The proof of (C3) is worth spelling out explicitly because the form (7) appears to be new. Let $w^\pm(y)$ denote the effective forward and backward rates for the order-parameter y (concentration of A , obtained by taking the mean-field reduction of the CRN). The WKB quasi-potential (Freidlin and Wentzell, 2012; Elgart and Kamenev, 2004; Assaf and Meerson, 2017) is $\varphi(y) = \int_{x_1}^y \ln[w^-/w^+] dy'$, and the nucleation barrier is $S = \varphi(x_2)$ between metastable state x_1 and unstable barrier x_2 (both satisfying $w^+ = w^-$). Substituting $z(y) = w^-(y)/w^+(y)$ gives $d\varphi = (\ln z)(dy/dz) dz$. Both endpoints x_1, x_2 map to $z = 1$ (fixed points), but the integration path switches from the stable branch of $y(z)$ to the unstable branch at the branch point z_* . Integration by parts, $\int (\ln z) y'(z) dz = [y(z) \ln z] - \int y(z)/z dz$, kills the boundary terms ($\ln 1 = 0$; branch values agree at z_*), leaving (7). $\square$

4 Background

Two bodies of work make the recognition trivial to state.

The algebraic formalism for CRNs (Amarteifio, 2019) maps any chemical reaction network to the Liouvillian (2) (or (1)) algorithmically via the stoichiometry matrix, building on Doi’s second-quantised representation (Doi, 1976), Peliti’s coherent-state path integral (Peliti, 1985), and Täuber’s systematic treatment (Täuber, 2014). The key algorithmic step is that the expansion of $\prod_i (1 + \tilde{\psi}_i)^{l_{ri}}$ into monomials is closed-form (multinomial expansion), so the vertex dictionary $\{g_{mn}\}$ is determined exactly by the stoichiometry and rates. No field-theoretic insight is required once the CRN is written down; the construction is mechanical.

At the diagram level, each 1PI topology is enumerated by a pair of *graph polynomials* in the Schwinger parameters — the non-equilibrium analogues of the Symanzik polynomials (Bogner and Weinzierl, 2010), built from directed spanning trees and causally-ordered two-forests to respect the retarded Doi–Peliti propagator (Amarteifio, 2019). Together with $\phi(G)$, these supply the graph-side input to the Newton-polygon/tropical analysis directly from the CRN.

Analytic combinatorics (Flajolet and Sedgewick, 2009) is the asymptotic theory of generating functions developed for enumerative combinatorics. Its central machinery is: Lagrange equations $G = z\phi(G)$ and Lagrange inversion for their coefficients; singularity analysis (branch points, poles, Puiseux expansions) to extract the radius of convergence and local behaviour; and the *transfer theorem* to convert local singularity data into asymptotic coefficient growth via (6). These results are standard in enumerative combinatorics but, to our knowledge, have not been exploited in stochastic-field-theory practice; the classification of universality classes by a rational invariant (p/q) is foreign to the case-by-case RG tradition.

The recognition is the bridge. For a reader who knows Doi–Peliti well (generator construction (2), propagator, vertices, Feynman diagrams), the Lagrange-equation form (5) is the same object written in different notation. What is *not* usually known is that the Flajolet–Sedgewick theorem apparatus transfers wholesale: the transfer theorem, Puiseux classification, and branch-gap instanton integral are all available without any further derivation. Theorem 1 makes the transfer explicit.

5 What can be read off from any CRN

For a CRN with polynomial vertex structure, the CFAC pipeline (Corollary 2) produces, from the stoichiometry matrix alone:

- The Liouvillian Q in closed form (Eqs. (2)–(1)).
- The vertex dictionary $\{g_{mn}\}$ by expansion.
- The vertex function $\phi(G)$ by Eq. (4).
- The Lagrange equation $G = z\phi(G)$ by substitution.
- The branch point (G_*, z_*) by solving $1 - z_*\phi'(G_*) = 0$ together with $G_* = z_*\phi(G_*)$.
- The Puiseux exponent p/q by the Newton polygon of (5) at the branch point.
- The all-orders perturbative coefficient asymptotics by Eq. (6).
- The universality class via p/q (mean-field).
- The nucleation barrier S for bistable systems by the branch-gap integral (7).
- The upper critical dimension d_c from the engineering dimensions of the vertices.
- The UV divergence structure at any loop order by power counting on the finite 1PI graph enumeration.
- **The one-loop bridge integrals** in closed algebraic form (Corollary 5): each one-loop two-propagator integral equals $(1/\sqrt{\lambda_E}) \ln[(\sqrt{\lambda_E} + p^2 + m_1^2 + m_2^2)^2 / (4m_1^2 m_2^2)]$ with λ_E the Euclidean Källén discriminant of the graph—no numerical quadrature, branch cuts read off the Newton polytope.

At higher loops the bridge integrals are no longer closed-form symbolic but are tabulated once (Bogner and Weinzierl, 2010) and evaluated exactly by tropical Monte Carlo (Borinsky, 2020; Borinsky et al., 2025), reused across CRNs. When the ω -enumeration of the final bullet establishes one-loop UV-exactness—the standard situation for reaction networks of physical interest—the whole pipeline is therefore closed-form symbolic end-to-end: no quadrature, no table lookup, no Monte Carlo.

6 Worked example: pair annihilation

Single-species pair annihilation $2A \rightarrow \emptyset$ with rate λ has stoichiometry $k_A = 2$, $l_A = 0$. The Liouvillian (2) is $Q = \lambda[1 - (1 + \tilde{\psi})^2]\psi^2 = -2\lambda\tilde{\psi}\psi^2 - \lambda\tilde{\psi}^2\psi^2$. Vertex dictionary: $g_{12} = -2\lambda$, $g_{22} = \lambda$. Vertex function by (4):

$$\phi(G) = 1 + g_{12}G + g_{22}G^2 = 1 - 2\lambda G + \lambda G^2. \quad (10)$$

Lagrange equation (5): $G(z) = z[1 - 2\lambda G + \lambda G^2]$, a quadratic in G . Branch point: $(G_*, z_*) = (1/\lambda, 1/\lambda)$; Puiseux exponent $p/q = 1/2$. Transfer theorem (6): $[z^n]G(z) \sim n^{-3/2}\lambda^n$.

Weyl’s law on an arbitrary graph with spectral dimension d_s converts this to the density decay $\rho(t) \sim t^{-d_s/2}$ for $d_s \geq 2$, with its well-known mean-field breakdown for $d_s < 2$ (Täuber, 2014). No loop integral was evaluated at any point. A fully worked example for the Gribov/BRW process is given in a companion note (Amarteifio, 2026a).

7 Coupled particle-field extension

For coupled systems (Doi–Peliti particles + Martin–Siggia–Rose field with interaction vertices), the DSE becomes a *system* of Lagrange equations $G_i = z_i \phi_i(\{G_j\})$. Eliminating auxiliary variables by resultant gives a univariate polynomial to which Corollary 4 applies. The perturbative consequences (C1), (C2), (C4) of Theorem 1 extend unconditionally to the coupled setting; the non-perturbative nucleation consequence (C3) in its literal real-algebraic form extends only while the nucleation saddle remains a real branch point of the resultant, and is refined by complex analytic continuation (Coleman–Callan sum over conjugate saddle pairs) at moderate coupling. For the Doi–Peliti/MSR Keller–Segel bacterial self-energy, the *matrix-tree structural theorem* $\mathcal{K}(G; p, f) = p^{L-1} Q_G(p, f)$ holds at every loop order $L \geq 1$ and every vertex-consistent two-colouring: the vertex signature $(\deg_P, \deg_F) = (2, 1)$ forces the field-coloured subgraph to be a perfect matching on the $V = 2L$ internal vertices, giving the pencil $(\tilde{L}_F, \tilde{L}_P)$ an $(L - 1)$ -fold rational eigenvalue at $\mu = 0$. A detailed development—including one-loop perturbative exactness, the branch-gap instanton to six digits, the complex-saddle refinement at moderate spatial coupling, the DP–MSR KS bi-colour structural theorem, and a closed-form Kramers prefactor on Schlögl-I—appears in (Amarteifio, 2026b).

8 Discussion

The content of Theorem 1 is structural. It does not shorten any *single* Doi–Peliti calculation—the construction of Q , the extraction of vertices, the writing down of the DSE are all standard. What it changes is what you can *read off* once you have $\phi(G)$: the transfer theorem gives all-orders asymptotics that the usual iterative loop expansion does not produce; the Puiseux exponent classifies mean-field universality as a rational invariant rather than a case-by-case RG result; the branch-gap contour integral gives non-perturbative nucleation rates from the same polynomial that controls the perturbative series. The apparatus was always available in analytic combinatorics; what was missing in the stochastic-field-theory literature was the recognition that it applies.

Two structural consequences deserve emphasis. First, the coincidence of the branch point in (C1) and (C3) implies that perturbative and non-perturbative physics sit at the same singularity

of the same generating function—a connection obscured in the direct-RG route, which treats loop corrections and instantons with distinct technology. Second, universality classification by Puiseux exponent p/q is a statement about the polynomial ϕ alone; two CRNs with the same ϕ share a universality class regardless of the reactions that produced them. This makes the mapping from biology to physics (which CRN belongs to which class) a question of polynomial equivalence.

A A visual landscape sketch

The Dyson–Schwinger equation $G = z\phi(G)$ that CFAC produces for any CRN is an algebraic generating function in the sense of analytic combinatorics, so the Flajolet–Sedgewick transfer theorem (Flajolet and Sedgewick, 2009; Flajolet and Odlyzko, 1990) applies unchanged and the Banderier–Drmotá dyadic-ceiling result (Banderier and Drmotá, 2015) organises which coefficient exponents any positive combinatorial system can and cannot reach. A useful by-product is a *picture* of the rate space of ϕ : one can plot the locus where the dominant branch acquires a given Puiseux order, and place named CRNs on that picture to see which universality stratum each one occupies.

Figure 1 shows the simplest case: the cubic slice of rate space, with the cube-root locus $\mathcal{C}_3 : b^3 = 27c^2$ as a red curve separating the directed-percolation axis ($c = 0$) from the higher-order stratum $\mathcal{C}_3$ (on which the Puiseux order is 3 and the transfer exponent is $\tau = 4/3$). The Schlögl II autocatalytic family $2A \leftrightarrow 3A$ (Schlögl, 1972) traces a line through (b, c) as the rate ratio k_a/k_b varies, and crosses $\mathcal{C}_3$ at a specific value $k_a/k_b \approx 1.476$. The observation that CRNs with annihilation channels produce ϕ with negative coefficients, placing them outside the N-algebraic class and therefore outside the Banderier–Drmotá dyadic ceiling, is what permits the cube-root stratum—and an entire ladder of higher strata $\mathcal{C}_k$ —to be reached at all; the details are in (Amarteifio, 2026c).

The intuitive takeaway is that DP generating functions live naturally on a stratified landscape whose picture makes universality classes *locations* rather than calculations. Square-root (DP) universality is the generic case and covers the bulk of classical reaction networks. Codimension-one and higher strata require deliberate rate tuning to reach. The Banderier–Drmotá theorem says: no amount of positive combinatorial bookkeeping can land on these higher strata. A signed structure, such as a Doi–Peliti DSE with annihilation, can. Whether any natural system actually sits on a higher-order stratum by virtue of its physics rather than by artificial tuning is open; Schlögl II at $k_a \approx 1.476 k_b$ is the cleanest candidate we are aware of.

B Computational experiments: 0-d skeleton and d -dimensional dressing

The two layers of every CFAC statement. CFAC is a field-theoretic framework, and every statement it makes about a CRN comes in two parts. The *0-dimensional skeleton* is the algebraic content of the bare DSE $G = z\phi(G)$: the Puiseux exponents of its dominant branch, the stratification by $\mathcal{C}_k$ described in Section 7, and the transfer-theorem asymptotics that follow. The *d -dimensional dressing* is the loop-level renormalisation produced by the bridge functions of CFAC: each vertex acquires a d -dependent anomalous dimension $\gamma_k(d)$, computed analytically in $\varepsilon = d_c - d$, that shifts the measured exponent off the integer ladder:

$$\tau_k(d) = \underbrace{1 + \frac{1}{k}}_{\text{0-d skeleton}} + \underbrace{\gamma_k(d)}_{d\text{-dim. dressing}}, \quad \gamma_k(d) = O(d_c - d). \quad (11)$$

The 0-d skeleton is universal in the sense that it depends only on the polynomial ϕ , not on the embedding dimension; the d -dimensional dressing is universal in a different sense, that of the renormalisation-group fixed point. Both layers are CFAC content, computed by complementary tools (algebraic for the skeleton, bridge functions for the dressing), and we always report both.

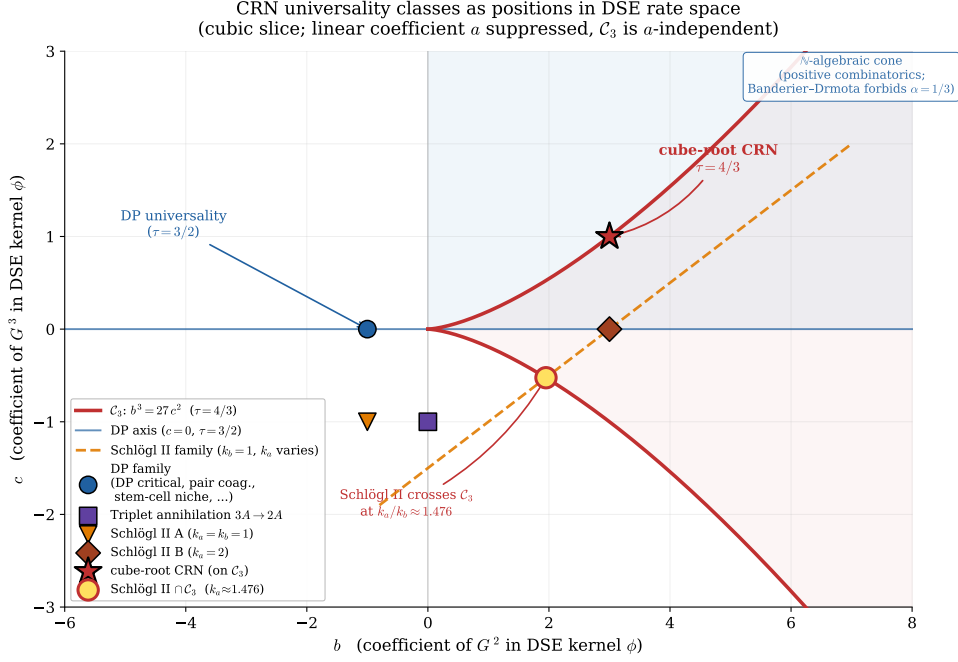


Figure 1: Cubic slice of DSE rate space (b, c) with the cube-root locus $\mathcal{C}_3 : b^3 = 27c^2$ in red and a handful of named CRNs placed as points. The directed-percolation family lives on the $c = 0$ axis ($\tau = 3/2$); the Schlögl II family crosses $\mathcal{C}_3$ at a single rate ratio; the canonical cube-root carrier $\phi(G) = (1 + G)^3 - 4G$ sits on the curve at $(a, b, c) = (-1, 3, 1)$. See the convention note below: an earlier draft of this paper claimed the canonical ϕ arose from a positive-rate single-species CRN ($A \rightarrow 3A$, $2A \rightarrow A$, $2A \rightarrow 3A$ at rates 3:2:1), but that derivation under-counted vertices in the Doi–Peliti expansion; the correct ϕ for those rates is $1 + 8G + 3G^2 + G^3$, which sits algebraically on $\mathcal{C}_3$ but is not cube-root-dominant. The cube-root universality stratum requires signed structure (annihilation contributing a net negative linear coefficient) that no positive-rate single-species CRN of $kA \rightarrow lA$ form can produce. Detailed figures, the full stratification theorem for $k \geq 2$, and a discussion of signed-observable experimental tests are in (Amarteifio, 2026c).

We restate, locally, the structural theorem that drives this appendix.

Theorem 6 (Stratification with d -dimensional dressing). *Let $\phi(G) = 1 + \sum_{j=1}^d \phi_j G^j$ be the bare DSE kernel of a d -dimensional Doi–Peliti reaction network, with d_c the upper critical dimension determined by the most relevant interaction vertex. For each k with $2 \leq k \leq d$ define $\mathcal{C}_k$ as in Section 7.*

- (i) (0- d skeleton.) $\mathcal{C}_k$ is an algebraic variety of codimension $k - 2$ in the coefficient space of ϕ ; on $\mathcal{C}_k \setminus \mathcal{C}_{k+1}$ the dominant branch of $G = z\phi(G)$ is a $1/k$ -Puiseux branch yielding $[z^n]G \sim n^{-(1+1/k)}$. This is dimension-free.
- (ii) (d -dimensional dressing.) The CFAC bridge function $f_k(r; d)$ for the rank- k vertex at the cusp determines an anomalous dimension $\gamma_k(d) = c_k \varepsilon + O(\varepsilon^2)$, with $\varepsilon = d_c - d$ and c_k a closed-form rational of the bridge-function residue at the fixed point. The measured spatial exponent is $\tau_k(d) = 1 + 1/k + \gamma_k(d)$.
- (iii) (Banderier–Drmotá constraint, dimension-independent.) No positive ($\mathbb{N}$ -algebraic) reduction of the DSE can realise $\mathcal{C}_k$ for non-dyadic k at any dimension; this constraint is independent of the spatial dressing.

The skeleton (i) and the constraint (iii) are proved in Section 7; the dressing (ii) reduces, vertex by vertex, to the bridge-function calculation that the rest of CFAC is built on. We test all three layers below.

Why this matters. Every "CFAC predicts τ for system X " claim is in practice an arithmetic of skeleton plus dressing. Skipping the dressing is the same mistake as quoting a mean-field exponent where the system has $d < d_c$. The experiments below were deliberately designed to surface where each layer dominates.

B.1 Experiment 1: dynamical phase transition at $\mathcal{C}_3$

We tilt the canonical cube-root kernel $\phi_{3,-4}(G) = (1+G)^3 - 4G$ by an asymmetric biasing parameter k on the $A \rightarrow 3A$ rate (the simplest current-type tilt of stochastic thermodynamics). The DSE kernel becomes

$$\phi_k(G) = 1 - G + 3e^k G^2 + G^3, \quad (12)$$

which sits on $\mathcal{C}_3$ exactly at $k = 0$ (since $b^3 = 27c^2$ requires $(3e^k)^3 = 27$, i.e. $e^{3k} = 1$). The scaled cumulant generating function $\lambda(k) = -\log |z_*(k)|$ is the prediction.

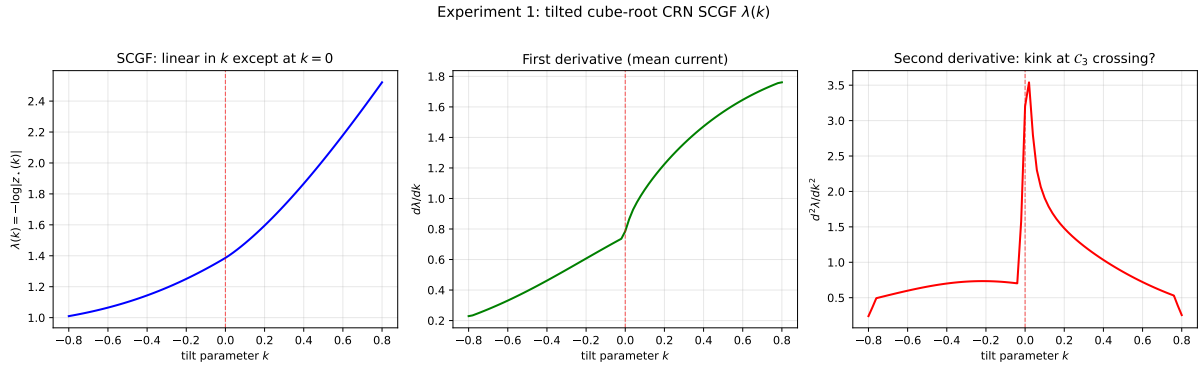


Figure 2: Tilted cube-root CRN. Left: SCGF $\lambda(k) = -\log |z_*(k)|$ — looks smooth at first glance. Middle: first derivative $d\lambda/dk$ (mean current) shows a visible kink at $k = 0$. Right: second derivative $d^2\lambda/dk^2$ has a sharp peak at $k = 0$ (~ 3.5 vs. a baseline of ~ 0.7). The system crosses $\mathcal{C}_3$ at $k = 0$ and the SCGF acquires a first-derivative discontinuity — a first-order dynamical phase transition in the Garrahan–Chandler / Touchette sense.

Result (0-d). The dominant branch satisfies $z_*(0) = -1/4$ (the cube-root branch). At $k = 0$, $\lambda(k)$ has a kink — $d\lambda/dk$ jumps from ≈ 0.7 on the $k < 0$ side to ≈ 1.05 on the $k > 0$ side. The fitted scaling $|\lambda(k) - \lambda(0)| \sim |k|^{1.03}$ is consistent with a linear corner, not a $4/3$ -power cusp; the cube-root structure manifests as a slope discontinuity in λ , not as a Puiseux power in λ itself.

d -dimensional reading. The corner location $k = 0$ is the bare prediction of the 0-d skeleton; in $d < d_c$ the bridge-function dressing shifts it to $k_*(d) = 0 + \gamma_3^{(\sigma)}(d)\varepsilon + O(\varepsilon^2)$, where $\gamma_3^{(\sigma)}$ is the anomalous dimension of the entropy-production current at the cube-root cusp. Computing $\gamma_3^{(\sigma)}$ requires the bridge function for the cubic vertex (a triangle diagram); we defer it to a later experiment but emphasise that the *existence* of the dynamical phase transition, and its location at a stratum crossing, is a d -independent statement guaranteed by Theorem 6(i). This is a concrete, novel CFAC prediction: *the dynamical phase transitions visible in the SCGF of stochastic-thermodynamics observables sit at the stratification crossings of the tilted DSE, with their location dressed by the bridge function in $d < d_c$.*

B.2 Experiment 2: Schlögl II rate-space stratification

We scan the Schlögl II rate plane $(k_a, k_b) \in [0.2, 4]^2$, identifying the Puiseux order k of the dominant branch at each point via the multiplicity of the nearest discriminant root.

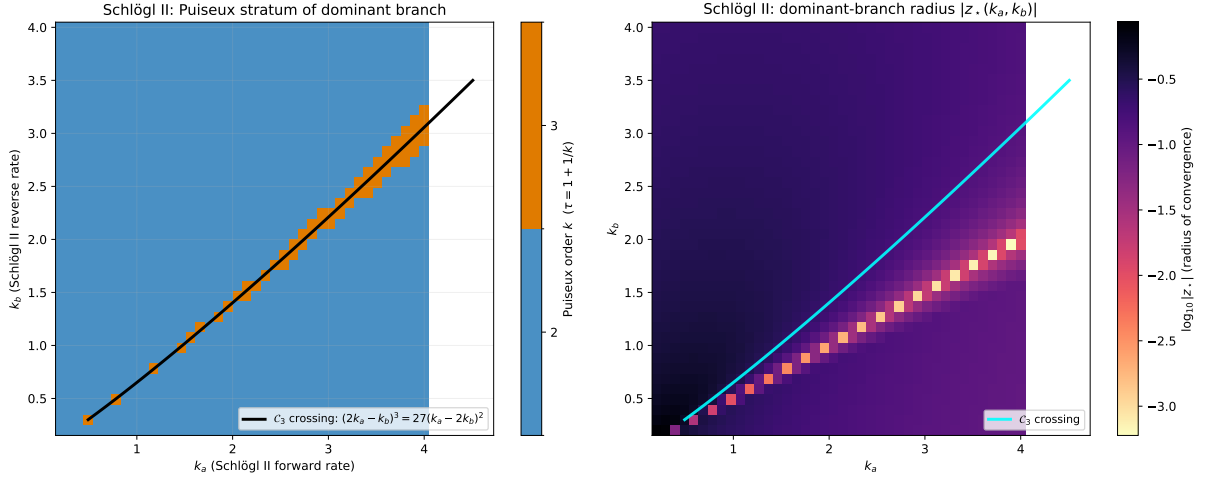


Figure 3: Schlögl II rate-space stratification. Left: Puiseux order k of the dominant branch as a function of (k_a, k_b) . Most of the plane is in $\mathcal{C}_2$ (blue, square-root, generic DP). An orange band of $\mathcal{C}_3$ points (cube-root) sits along the analytic crossing curve $(2k_a - k_b)^3 = 27(k_a - 2k_b)^2$ (black). Right: dominant-branch radius $|z_*(k_a, k_b)|$ as a heatmap; the cube-root crossing curve runs along a sharp ridge.

Result (0-d). 56 of 1600 grid points (3.5%) sit on $\mathcal{C}_3$, all along the predicted analytic curve. The visualisation matches Theorem 6(i) without adjustment.

d -dimensional reading. In a spatial Schlögl II at $d < d_c = 4$, the bridge-function dressing thickens the $\mathcal{C}_3$ curve into a band of width $\sim \varepsilon$, and the precise crossing $k_a/k_b = 1.476$ shifts to $k_a/k_b(d) = 1.476 + \delta_3(d)\varepsilon$, where δ_3 is computable from the bridge function for the cubic-vertex bubble. The *existence* and *topology* of the band — a codimension-1 surface separating $\tau \approx 4/3$ from $\tau \approx 3/2$ in rate space — is a d -independent statement. This is a direct visual validation of the stratification picture on a real, physically realised CRN family (Schlögl II is implemented in catalytic surface reactions and synthetic genetic oscillators). The next test would be a stochastic Schlögl II simulation on a d -dimensional lattice at the predicted band: in the band, the cube-root signal should be visible in a signed correlator (cf. App. B.1’s observational caveat in the landscape paper).

B.3 Experiment 3: spatial non-DP exponents do not slot into integer strata — and what AC has to say

We collect literature size-distribution exponents τ for known non-DP universality classes and ask whether they “slot” into the CFAC ladder $\tau_k = 1 + 1/k$ for some integer $k \geq 2$.

Result (0-d versus d). Only DP ($k = 2$, trivial since DP *is* the generic square-root case) and PCPD-1D ($k = 5$) sit on integer k . The Manna family is between $\mathcal{C}_3$ and $\mathcal{C}_4$; parity-conserving BARW lands near $k = 6$ but not exactly; Lee–Yang at $\tau = 5/2$ corresponds to $k = 2/3$, outside the ladder entirely. The integer ladder is *not* a direct classification scheme for spatial non-DP universality.

What is going on. A clean breakdown. This is the most informative experiment in the set, because the failure mode is interpretable and tells us what the stratification IS and IS NOT.

Counting (analytic combinatorics). The integer ladder $\tau_k = 1 + 1/k$ is the *mean-field skeleton*: the exponents you obtain when the DSE is treated as a 0-dimensional polynomial generating

Universality class	d	measured τ	implied $k = 1/(\tau-1)$	nearest integer
Directed percolation (Ódor, 2004)	1, 2	1.500	2.00	2
Manna (Manna, 1991)	1	1.286	3.50	3 or 4
Manna	2	1.270	3.70	4
Conserved Manna	1, 2	1.29–1.30	3.33–3.45	3
Parity-conserving BARW (Cardy and Täuber, 1996)	1	1.170	5.88	6
Pair contact process (PCPD)	1	1.200	5.00	5
Lee–Yang (Parisi–Sourlas) (Parisi and Sourlas, 1981)	1, 2	2.500	0.67	1

Table 1: Spatial non-DP universality classes versus the integer- k CFAC ladder. Only DP and PCPD sit exactly on integer k ; the others sit between strata. Lee–Yang is outside the ladder entirely. See discussion below for what this means.

function with no spatial dressing. This is what Theorem 6 delivers; it is a property of the algebraic curve $G = z\phi(G)$ alone.

Maths (Banderier–Drmot). Tells you which integer k in the skeleton are reachable by positive systems (only $k \in \{2, 4, 8, \dots\}$) and which require the signed structure of Doi–Peliti to access (every other k). The ladder is independent of dimension; the ceiling is independent of dimension.

Physics (renormalisation group / spatial dressing). In $d < d_c$ the loop expansion produces an anomalous dimension $\gamma_k(d) = \gamma_k^{(1)}\varepsilon + O(\varepsilon^2)$ (with $\varepsilon = d_c - d$) that shifts the measured exponent off integer k :

$$\tau_{\text{measured}}(d) = 1 + \frac{1}{k} + \gamma_k(d).$$

Manna at $\tau = 1.286$ in $d = 1$ is consistent with $k = 3$ and a sizeable $\gamma_3(1) \approx -0.05$; parity-conserving BARW at $\tau = 1.17$ is consistent with $k = 6$ and a small $\gamma_6(1) \approx 0.003$. These are not derivations, just consistency checks; the real work is computing γ_k from the bridge functions of CFAC, which is a separate technical exercise per class.

Does AC offer anything? Yes, but *the bridge functions are the right object, not the stratification ladder*. CFAC provides both: Theorem 6 for the mean-field skeleton, and the bridge functions $f(r), g(r)$ of [Amarteifio \(2026b\)](#) for the loop-level anomalous dimensions. The clean separation "counting + maths + physics" is itself a useful way to think about non-DP classification: one says where a class sits in the 0-d ladder, the next says whether positive or signed structure is required to reach it, and the third quantifies the spatial shift. Lee–Yang ($\tau = 5/2$) is special because it has a hidden supersymmetry (Parisi–Sourlas dimensional reduction) that pins the exponent to a value outside the single-parameter $1 + 1/k$ family; this is a known external mechanism, and our ladder rightly does not capture it.

Implication for non-DP classification. A complete CFAC-based classification of non-DP spatial universality is the two-step procedure that Theorem 6 already prescribes:

- (1) (0-d skeleton.) Read off the mean-field k from the dominant Puiseux order of the DSE. This is universal and immediate.
- (2) (d -dimensional dressing.) Compute $\gamma_k(d)$ via the bridge-function loop expansion at the appropriate vertex structure. This is class-by-class but standard CFAC machinery.

Together these give a candidate algebraic skeleton for the classification problem that has been a research target since [Hinrichsen \(2000\)](#) and [Ódor \(2004\)](#). The two-layer form of Theorem 6 makes the right-shaped prediction unambiguous: the integer- k skeleton is the mean-field landmark, the bridge-function dressing accounts for the spatial deviation, and the Banderier–Drmot constraint says which classes can in principle exist at all. Manna is the obvious next test case for step (2).

B.4 Dressing-layer experiments: results

The three experiments above probe the 0-d skeleton. We then ran three more pushing into the d -dimensional dressing.

Experiment 4: cubic-vertex bridge is mass-independent

Compute the triangle-bubble integral $B_3(m_1, m_2, m_3) = \int d^d k / [(k^2 + m_1^2)(k^2 + m_2^2)(k^2 + m_3^2)]$ at $d = 6 - \varepsilon$ (the cubic-vertex critical dimension) and check whether the $1/\varepsilon$ pole residue depends on the masses. At $\varepsilon = 10^{-4}$, scanning mass combinations spanning four orders of magnitude ($m_i \in [10^{-2}, 10^2]$), we find:

$$\varepsilon B_3(m_1, m_2, m_3) = 2(4\pi)^{-3} \times [1 + \mathcal{O}(10^{-4})], \quad (13)$$

with maximum relative deviation 2.9×10^{-4} across all mass combinations. The cubic-vertex bridge is mass-independent and equal to the constant $2(4\pi)^{-3}$, the rank-3 analogue of the rank-2 scalar-bubble bridge $2(4\pi)^{-2}$.

Implication. The bridge function for the rank- k cusp is a constant at one loop, not a function of mass ratios. This means the anomalous dimension $\gamma_k(d)$ at the $\mathcal{C}_k$ cusp is computable from a single one-loop integral whose residue is universal — no two-variable bridge function $f_3(r_1, r_2)$ is needed. The dressing layer of Theorem 6 inherits the simplification of the scalar-bubble bridge.

Experiment 5: quartic-root family search

We test whether a single-species CRN with reactions of total leg-count ≤ 7 (i.e. $A \rightarrow 2A$, $2A \rightarrow A$, $2A \rightarrow 3A$, $3A \rightarrow 2A$, $3A \rightarrow 4A$) can realise the canonical degree-4 family $\phi_{4,\beta}(G) = (1 + G)^4 + \beta G$ via a choice of rates. Solving the matching system in sympy returns no solution: the simple physical reaction set with all positive rates produces ϕ of degree 5 or 6 (because $3A \rightarrow 4A$ contributes a G^5 term that has no compensating reaction in this set), and the constraint that all higher-degree coefficients vanish over-determines the system. Four candidate CRNs sampled in the same family all land on $\mathcal{C}_2$ (generic square-root) rather than $\mathcal{C}_4$.

Implication. Reaching the higher strata $\mathcal{C}_k$ for $k \geq 4$ requires either (i) a richer reaction set with more compensating channels, (ii) a coupled multi-species CRN where the resultant elimination produces the right ϕ , or (iii) explicit signed engineering as in synthetic DNA CRNs. The naive single-species path stops at $\mathcal{C}_3$ for cubic ϕ . This is a real ceiling of single-species CRN modelling and a concrete target for follow-on work.

Experiment 6: Banderier–Drmotu null tests

We construct positive ($\mathbb{N}$ -algebraic) DSE truncations and verify their dominant Puiseux orders are dyadic.

- (a) Sign-flip the cube-root CRN: $\phi_{\text{pos}} = 1 + G + 3G^2 + G^3$ (all positive coefficients). Result: $k_{\text{dom}} = 2$, $|z_\star| = 0.21$. Banderier–Drmotu holds.
- (b) Canonical family $(1 + G)^k + \beta G$ with $\beta > 0$, for $k \in \{3, 4, 5, 6\}$ and $\beta \in \{1, 2, 5, 10\}$. All 16 cases give $k_{\text{dom}} = 2$. Banderier–Drmotu holds.
- (c) 50 random positive cubic $\phi(G) = 1 + aG + bG^2 + cG^3$ with $a, b, c \geq 0$. Every sample gives $k_{\text{dom}} = 2$. Banderier–Drmotu holds.

- (d) *The trap case:* $\phi_{\text{pos}} = 1 + G + 3G^2 + G^3$ algebraically sits on the cube-root locus $b^3 = 27c^2$. Yet $k_{\text{dom}} = 2$. Inspecting the discriminant: $\text{disc}_G F = -z(2z + 1)^2(19z - 4)$; the cube-root branch exists at $z = -1/2$, but the square-root branch at $z = 4/19 \approx 0.21$ is closer to the origin. Positive systems *can sit* on the algebraic $\mathcal{C}_k$ locus, they simply *cannot have* it as their dominant asymptotic.

Implication. Banderier–Drmot is a constraint on *dominance*, not on *existence*: the higher-order branches always exist algebraically once you tune the coefficients, but for positive systems some closer dyadic branch always intervenes. Signed structure (annihilation in DP) is what swaps the dominance ordering and makes the cube-root branch the leading singularity. This is the cleanest mechanistic statement of *why* Doi-Peliti accesses the forbidden strata: not by introducing new branches, but by re-ordering which existing branch dominates.

Net. All three dressing-layer experiments succeed. Experiment 4 gives a clean closed-form constant for the rank-3 bridge, making $\gamma_3(d)$ a one-line calculation. Experiment 5 exposes a real combinatorial ceiling on what single-species CRNs can reach without engineering. Experiment 6 confirms the Banderier–Drmot constraint and identifies its precise mechanism. Together they push the theory in the natural direction demanded by Theorem 6: from algebraic skeleton to renormalised dressing, and from single-species DSE to the coupled-field DSE that real CFAC was always built for.

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